

Math 514, F19
Quiz 4

Name: Answer

Instructions: You must show all of your work to receive credits for correct answers.

Problem 1. (5 points) Let $S(n)$ be the price of a certain stock at the end of n weeks. Suppose that $S(n)/S(n-1)$ is lognormal with mean $\mu = 0.0006$ and standard deviation $\sigma = 0.035$. Estimate the probability that the stock price will be up by at least 5% at the end of 3 weeks.

$$\begin{aligned}
 P\left\{\frac{S(3)}{S(1)} > 1.05\right\} &= P\left\{\frac{S(3)}{S(2)} \cdot \frac{S(2)}{S(1)} \cdot \frac{S(1)}{S(0)} \geq 1.05\right\} \\
 &= P\left\{\log\left(\frac{S(3)}{S(2)}\right) + \log\left(\frac{S(2)}{S(1)}\right) + \log\left(\frac{S(1)}{S(0)}\right) \geq \log(1.05)\right\} \\
 &\quad \downarrow X_1 \quad \downarrow X_2 \quad \downarrow X_3 \\
 &\quad E[X_1 + X_2 + X_3] = 3\mu = 3 \times 0.0006 = 0.0018 \\
 &\quad \text{Var}[X_1 + X_2 + X_3] = 3\sigma^2 = 3(0.035)^2 \\
 &= P\left\{Z \geq \frac{\log(1.05) - 0.0018}{0.035 \cdot \sqrt{3}}\right\} \approx P\left\{Z \geq \frac{0.04699}{0.06062}\right\} \\
 &\approx P\{Z \geq 0.77516\} = 1 - P\{Z < 0.77516\} \\
 &= 1 - \Phi(0.77516) \approx 1 - 0.7809 \\
 &= 0.2191
 \end{aligned}$$

Problem 2. (5 points) Rolling a fair die 500 times. What is approximately the probability that the number 4 appears more than 100 times by using Central Limit Theorem?

Define $X_i = \begin{cases} 1 & \text{if the number 4 appears in the } i\text{-th rolling} \\ 0 & \text{otherwise} \end{cases}$

A Bernoulli
random variable
with $P = \frac{1}{6}$

clearly $E[X_i] = 1 \cdot \frac{1}{6} + 0 \cdot \frac{5}{6} = \frac{1}{6}$, $\text{Var}(X_i) = P(1-P) = \frac{1}{6} \cdot \frac{5}{6} = \frac{5}{36}$

Let $X = \sum_{i=1}^{500} X_i$, then $E[X] = \frac{1}{6} \cdot 500 = \frac{250}{3}$
 $\text{Var}(X) = 500 \cdot \frac{5}{36} = \frac{625}{9}$

Thus by CLT, we have.

$$\begin{aligned}
 P\{X > 100\} &\approx P\left\{Z > \frac{100 - \frac{250}{3}}{\sqrt{625/9}}\right\} = P\{Z > 2\} \\
 &= 1 - P\{Z \leq 2\} = 1 - \Phi(2) \approx 1 - 0.9772 = 0.0228
 \end{aligned}$$